UNIT - III

Liquid Penetrant, Eddy Current & Magnetic Particle

PART A - LIQUID PENETRANT TEST (LPT / PT)

1. Basic Concepts

* Liquid penetrant testing detects surface-breaking discontinuities (cracks, laps, seams, porosity, cold shuts) in non-porous materials: metals, plastics, ceramics, glass.
* A coloured or fluorescent liquid with very low surface tension enters the flaw by capillary action ; after removing the excess, a developer draws the trapped liquid back out (blotting) and makes a visible indication .
* Cannot find sub-surface flaws, and cannot be used on porous materials (powder metals, wood).

Physical Principles

* Surface tension: property of liquid surface to act like a stretched skin; penetrant has low surface tension so that it spreads and wets the surface.
* Wetting ability and contact angle: good penetrant wets the surface with contact angle less than 90 degrees (nearly 0).
* Capillary action: liquid rise in a narrow gap, $h = 2\,T\cos(\theta) / (\rho\, g\, r)$ - narrower crack = higher force.
* Viscosity: low viscosity -> fast entry, but too low drains out of flaw.
* Specific gravity, volatility, flash point, chemical inertness and toxicity are other properties checked.
* Sensitivity of penetrant depends on its ability to enter fine tight cracks and to give a bright indication.

> **Remember:**
> LPT works by capillary action in the entry stage and by reverse capillary / blotting action in the development stage.

2. Liquid Penetrant System

2.1 Types of Penetrants

* **Type I – Fluorescent penetrant:** glows (yellow-green) under **ultraviolet (black) light of 320 – 400 nm**, intensity 1000 microwatt per sq cm; very high sensitivity; needs darkened area.
* **Type II – Visible (colour contrast) penetrant:** usually **red dye** against white developer; viewed in normal white light (minimum 1000 lux); good for field work.
* **Type III – Dual sensitivity:** both visible and fluorescent.

2.2 Sensitivity Levels

* Level 1/2 – ultra low / low; Level 3 – high; Level 4 – ultra high sensitivity (for very fine cracks).

2.3 Methods of Removal of Excess Penetrant (Methods A, B, C, D)

Method	Type	Removal
A	Water washable	Washed with water spray
B	Post-emulsifiable, lipophilic	Emulsifier (oil based) then water wash
C	Solvent removable	Wipe with lint free cloth moistened with solvent
D	Post-emulsifiable, hydrophilic	Pre-rinse, water-based emulsifier, final rinse

2.4 Developers

* **Dry powder** – used mainly with fluorescent penetrant.
* **Water soluble and water suspendable** – powder in water, sprayed or dipped.
* **Non-aqueous wet developer** – powder suspended in solvent, sprayed (most common for visible penetrant).
* **Plastic film developer** – special applications.
* Developer action: acts like a **blotter**, spreads the bleed-out to increase visibility, gives white **contrast background**.

2.5 Other Materials

* **Cleaners / solvents** (pre-cleaning, post-cleaning), **emulsifiers**, **remover**; aerosol cans for field kits; UV lamp and radiometer; test panels (TAM, PSM) for checking the system performance.

3. Test Procedure - LPT

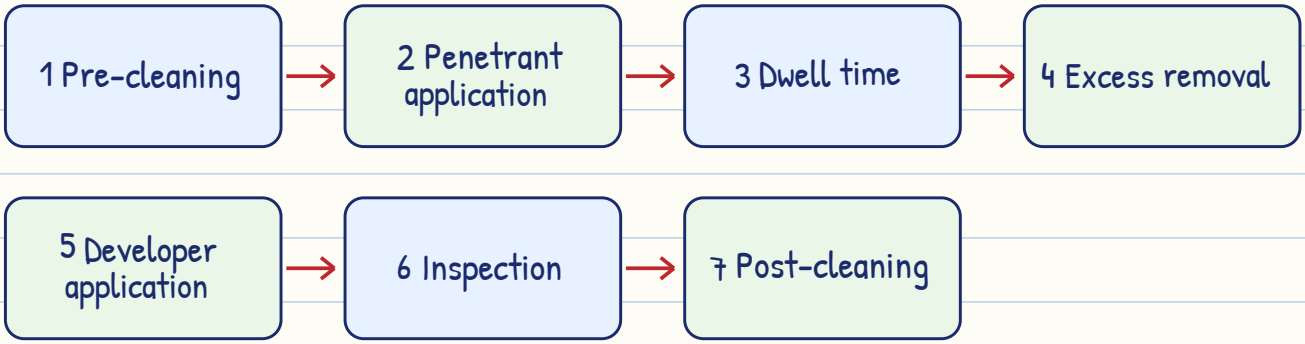


Fig: Steps of liquid penetrant testing

* **1. Surface preparation / pre-cleaning:** remove oil, grease, paint, rust, scale, dirt using solvent, vapour degreasing, detergent, ultrasonic cleaning, or wire brushing. **Do not use shot blasting** or heavy machining which smears metal over the opening of cracks. Dry the part completely.
* **2. Penetrant application:** spray, brush, dip or flood the surface (temperature 10 - 52 deg C).
* **3. Dwell time:** time the penetrant stays on the surface to enter flaws - **5 to 30 minutes** depending on material, flaw type, temperature (typical 10 - 20 min). Short for cracks in aluminium castings, longer for tight fatigue cracks.
* **4. Removal of excess penetrant:** as per method A to D; avoid over-washing (which removes penetrant from the flaw) and under-washing (leaving a high background).
* **5. Developer application:** thin uniform coat; **development time 7 - 30 minutes** (generally a minimum of 10 min).
* **6. Inspection:** under proper white light (for visible) or UV light in dark area; look for bleed-out within the development time. Flaws are **marked, recorded and measured** .
* **7. Post cleaning:** remove developer and penetrant residue to prevent corrosion.

Interpretation of Indications

Indication	Meaning
Continuous fine line	Crack, cold shut, seam
Intermittent line	Tight crack or crack partly closed
Round dots / clusters	Porosity, pin holes
Large diffuse patch	Excessive penetrant, rough surface; false indication
Wide bleed-out spread	Deep or wide flaw holds more penetrant

* **Relevant indication** from real flaws; **non-relevant indications** from part design (keyways, press fit, rough surface); **false indications** from poor processing, e.g. fingerprints, lint, penetrant on hands.
* Acceptance as per ASME Sec. V Art.6 / Sec. VIII Appendix 8, ASTM E165, E1417, IS 3658: **no linear indication longer than 1.5 mm** in most pressure vessel codes; rounded indications up to 5 mm; cracks never accepted.

4. Effectiveness and Limitations of LPT

* **Advantages:** simple, cheap, portable, high sensitivity to **fine surface cracks**, works on any non-porous shape and size, one-shot full surface coverage, easy to interpret, little training.
* **Limitations:** detects **only surface-opening flaws**; surface must be clean and smooth (not porous or very rough); not for painted / oxide covered surface; temperature range limit; time consuming steps; **no depth information**; chemical hazards and disposal of liquids; skin contact and fire hazard.

> **Tip:**
> To avoid missing tight cracks: more dwell time, ultra-high-sensitivity fluorescent penetrant, thin uniform developer and good pre-cleaning.

PART B - EDDY CURRENT TEST (ECT)

5. Principle of Eddy Current

* Based on **electro-magnetic induction (Faraday's law)**.
* An **alternating current in a coil** sets up an **alternating primary magnetic field**. When this coil is placed near a **conducting material**, circulating currents called

eddy currents are induced in it.

* Eddy currents set up a **secondary magnetic field** that opposes the primary field (Lenz's law), changing the **impedance of the test coil**.
* A **flaw, change in conductivity, permeability, dimension or heat treatment** disturbs the eddy currents -> the coil impedance (resistance and reactance) changes -> signal shown on the instrument.

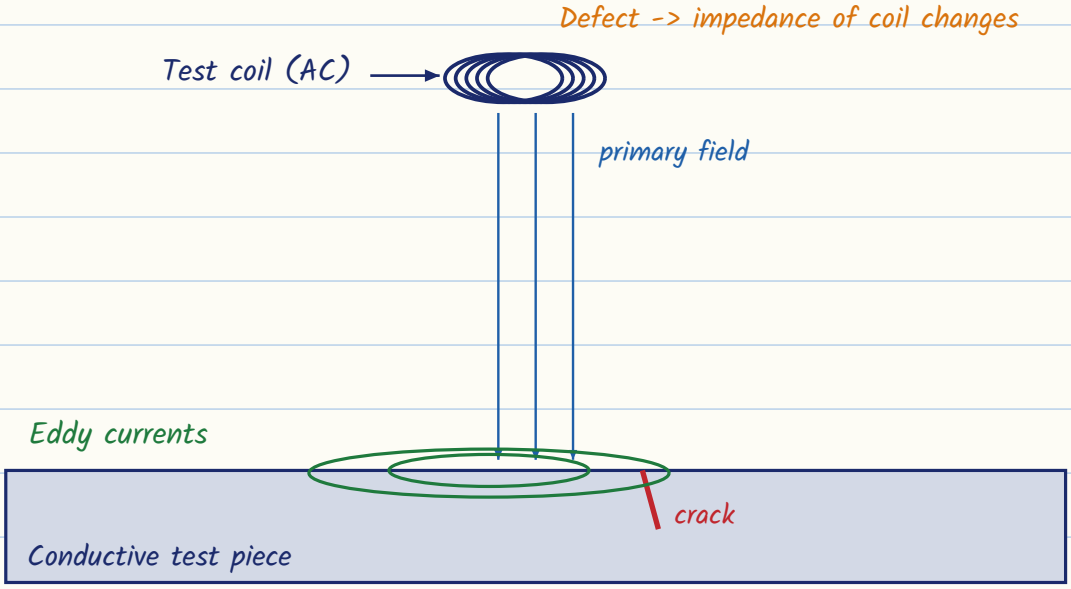


Fig: Principle of eddy current testing

Factors Affecting Eddy Currents

* **Electrical conductivity (sigma)** of the material.
* **Magnetic permeability (mu)** - ferromagnetic materials give strong noise.
* **Frequency of the test current (f)** - changes the depth of penetration.
* **Geometry** - thickness, curvature, edge effect.
* **Lift-off** - distance between the coil and the test surface.
* **Fill factor** - ratio of the specimen volume to coil volume (for encircling coils).
* **Presence of defects** - cracks interrupt the current flow.

Skin Effect and Depth of Penetration

* Eddy current density is **maximum at the surface** and falls exponentially with depth - the **skin effect**.
* **Standard depth of penetration: delta = 1 / sqrt(pi f mu sigma)** - the depth where the current density is 37% of the surface value.

* Higher frequency, conductivity or permeability -> **smaller depth of penetration** .
* Use **low frequency for sub-surface flaws** and high frequency for surface flaws; choose 3 delta as effective depth.

6. Eddy Current Test System

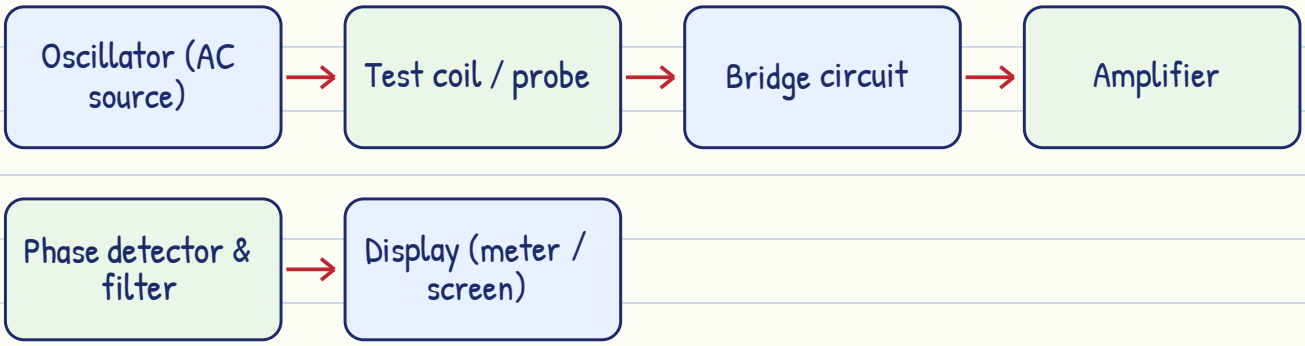


Fig: Block diagram of eddy current instrument

6.1 Probes and Coils

* **Encircling (feed-through) coil:** part passes through the coil - for tubes, wires, bars; fast testing.
* **Internal (bobbin) coil:** inserted in tubes - heat exchanger tube inspection.
* **Surface (pancake) probe:** flat coil on the surface - plates, welds, bolt holes.
* **Absolute coil:** single coil, measures total property - good for conductivity, thickness, slow changes.
* **Differential coil:** two coils compare adjacent areas - ignores gradual changes, sensitive to **small flaws** .
* **Reflection (driver-pickup) probes:** separate exciter and sensor coil.
* **Shielded probes:** ferrite shield narrows the field.

6.2 Instrument Display

* **Impedance plane diagram:** plots resistance (X-axis) against inductive reactance (Y-axis) - **phase angle indicates flaw depth** , amplitude the size; lift-off signal is separated by phase.
* Other: meter reading, alarm, strip chart, C-scan, digital data.
* Multi-frequency instruments remove unwanted signals (such as support plates).

6.3 Calibration

* Reference standard of same material with **artificial defects** – **EDM notches**, **drilled holes** of known size. Instrument gain and phase are adjusted; conductivity checked with IACS standards (International Annealed Copper Standard).

7. Applications of Eddy Current Testing

* **Flaw detection:** surface and sub-surface cracks, pits, corrosion in tubes (heat exchangers, boilers, condensers), bolt holes in aircraft structures, welds.
* **Conductivity measurement:** sorting of materials, checking **heat treatment**, alloy identification, hardness.
* **Thickness measurement:** thin metal sheets, foils, and **coating thickness** (non-conductive paint on metal).
* **Sorting** of mixed materials and of heat-treated parts.
* **Wire, rod and tube testing** in production lines at high speed.
* **Remote field ECT** for ferromagnetic tubes; **pulsed ECT** for insulated pipes; **array ECT** for fast scanning.
* **Aerospace** – fuselage lap joints, engine disks; railway wheel and rail inspection.

8. Effectiveness of Eddy Current Testing

* **Advantages:** no couplant or contact needed; **high speed** and easy automation; sensitive to **small surface cracks**; works through paint and thin coatings; portable; gives instant result; also for conductivity and thickness; works at **high temperatures**.
* **Limitations:** only for **electrically conductive materials**; shallow depth (skin effect); difficult for **ferromagnetic** materials (use saturating coil); complex signal interpretation; the **lift-off** and edge effect problems; flaws parallel to the coil winding direction may be missed; reference standards needed; geometry limits the probe selection.

Remember:

ECT = conducting material, AC coil, change in coil impedance. Depth falls as frequency increases.

PART C - MAGNETIC PARTICLE TEST (MPT / MT)

9. Magnetic Materials

* **Ferromagnetic materials** (iron, nickel, cobalt and most steels) are **strongly attracted** by magnet and can be magnetised easily; relative permeability is very high (100 to 100000).
* **Paramagnetic** (aluminium, platinum, magnesium) - weakly attracted, permeability slightly more than 1.
* **Diamagnetic** (copper, gold, silver, zinc, lead) - weakly repelled, permeability a little less than 1.
* MPT can be applied only to **ferromagnetic materials** .

Key Terms

* **Magnetic flux density B** (tesla), **field strength H** (A/m), **permeability mu = B / H** .
* **Retentivity:** ability to retain magnetism after the field is removed. **Coercive force:** reverse field needed to remove residual magnetism. **Reluctance** - opposition to flux.
* **Hysteresis loop (B-H curve):** hard magnets have a wide loop (high retentivity), soft magnets a narrow loop.

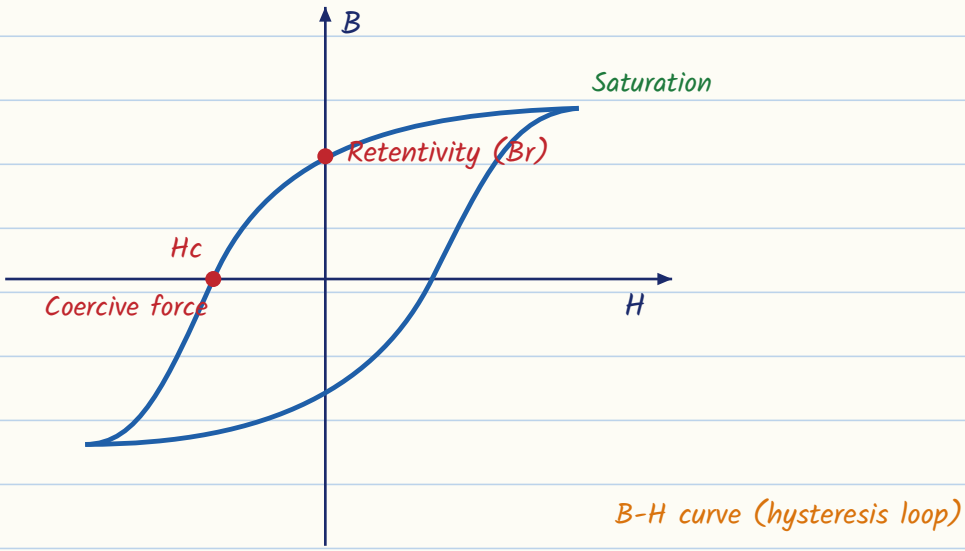


Fig: B-H curve of a ferromagnetic material

10. Magnetization of Materials

* A part is magnetised by passing **current through it** (circular field) or by **placing it in a magnetic field** (longitudinal field).
* The magnetic field must be **at 45 - 90 degrees to the flaw** for best indication. Hence two perpendicular fields are applied one after another.

Circular Magnetization

* Current passes through the part (head shot) or a central conductor (threader bar) -> circular field around the current path. Detects **longitudinal flaws** (parallel to the axis).
* **Prod method:** two hand-held electrodes (prods) pressed on the surface, current flows between them (spacing 75 - 200 mm; current 90 - 110 A per 25 mm spacing).

Longitudinal Magnetization

* Part placed in a **coil / solenoid** or between the poles of an **electromagnetic yoke** . Detects **transverse flaws** (across the axis).
* Coil: for length / diameter ratio (L/D) of 2 to 15, effective field strength depends on ampere-turns (NI).

Types of Magnetizing Current

* **AC (alternating current):** skin effect so **best for surface cracks** ; easy to demagnetise; used in portable yokes.
* **DC / rectified half wave (HWDC):** penetrates deeper, **subsurface flaws** , full wave (FWDC) smooth.
* **Pulsed / capacitor discharge** and **permanent magnets** are other sources.
* **Multi-directional magnetisation** with swinging fields for complex parts.

Methods of Magnetization

Method	Field	Flaw detected
Head shot / direct contact	Circular	Longitudinal flaws

Central conductor	Circular	Flaws on inside and outside surface of tubes
Prods	Circular	Weld flaws, large parts
Coil	Longitudinal	Transverse flaws
Yoke	Longitudinal (in between poles)	Any flaw perpendicular to the pole line, welds

II. Principle of Magnetic Particle Test

* When a ferromagnetic part is magnetised, magnetic flux lines flow through the part. A ***discontinuity (crack) perpendicular to the flux*** acts as an ***air gap of high reluctance*** , and the flux is forced out of the surface, forming a ***magnetic leakage field*** (a pair of poles).
* Fine ***ferromagnetic particles*** sprinkled on the surface are attracted to the leakage field and ***collect along the flaw*** , forming a visible indication that is wider and clearer than the flaw itself.

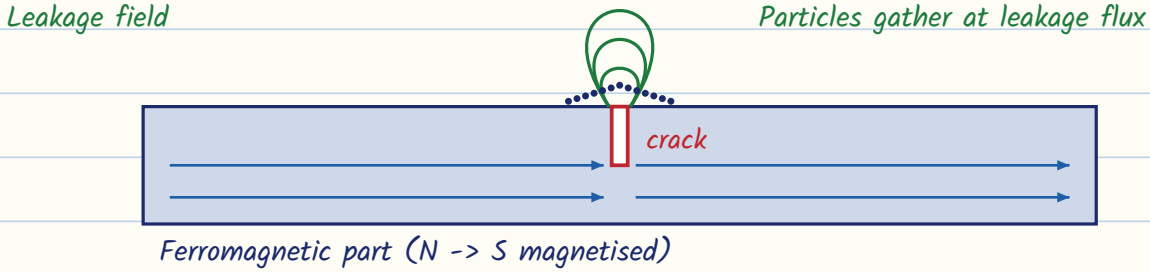


Fig: Leakage flux at a crack attracts magnetic particles

* Flaws perpendicular to the lines of flux give the best indication; ***parallel flaws are not detected*** .
* Sensitivity is highest for ***surface and near surface*** flaws (up to about 3 mm depth); decreases rapidly with depth.

Magnetic Particles

* **Dry powder:** iron oxide particles (50 – 150 micron) – coloured red, grey, black, yellow; good for rough surfaces, subsurface flaws with DC.
* **Wet particles:** fine particles (5 – 10 micron) in water or oil carrier – **higher sensitivity** for fine surface cracks, bath concentration 1.2 – 2.4 ml per 100 ml (visible), 0.1 – 0.4 ml (fluorescent).
* **Fluorescent particles** viewed under black light (UV) – best sensitivity.
* Visible particles are chosen to give **contrast** with the part (black on white lacquer background for wet).

12. Magnetic Particle Test Equipment

* **Portable electromagnetic yoke** (AC / DC) with articulated legs.
* **Prods and cable set. Permanent magnet yoke** (for places without power, explosive atmosphere).
* **Horizontal wet bench (stationary)** with head and tail stocks, coil, bath tank, pump, UV lamp and demagnetizing coil – for mass production.
* **Mobile units** , **flexible cable (coil wrapped) and threaders** .
* **Light meter and UV radiometer** , magnetic field indicator (pie gauge, Hall effect gaussmeter), and **Quantitative Quality Indicator (QQI)** shims, ASME **Ketos ring** to check system performance.

13. Magnetic Particle Test Procedure

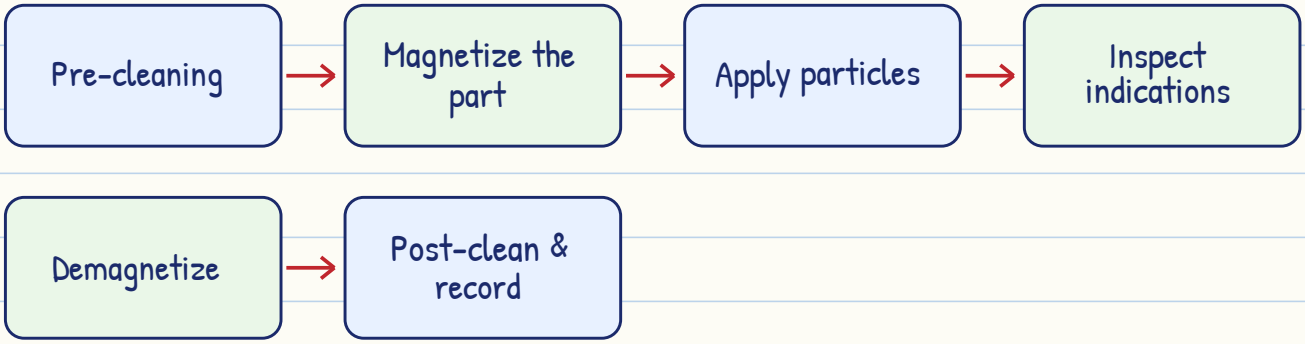


Fig: Magnetic particle testing steps

* **1. Preparation:** clean oil, scale, paint; surface within 1 mm coating is allowed with sensitivity check.
* **2. Magnetisation:** choose method and current (rule of thumb: 12 – 36 A per mm of diameter for head shot; yoke must lift 4.5 kg if AC and 18 kg if DC).

* **3. Application of particles: continuous method** (particles applied while the current flows - recommended, more sensitive) or **residual method** (applied after current is removed - for high-retentivity steels).
* **4. Inspection:** under white light (minimum 1000 lux) or UV light (1000 microwatt per sq cm in dark area at the surface); flaws form sharp, tightly held indications.
* **5. Demagnetisation** (below) and **6. post-cleaning and protection** against rust.

14. Demagnetization of Materials

* Part retains **residual magnetism** after testing. It can attract chips and swarf, damage bearings and gears, interfere with welding (arc blow), affect instruments, compass and subsequent machining.
* **Need for demagnetisation:** for moving parts, aircraft, welding next, plating, electronic instrument parts.
* **Method:** the part is subjected to an **alternating field of decreasing strength** , which reverses the domain direction in smaller and smaller steps until the residue is nearly zero.
  - **AC coil:** pass the part slowly through a coil with AC (about 5 - 7 A) and withdraw it 1 m or more away.
  - **DC reversing and decaying** for deep magnetised parts.
  - **Heating above Curie temperature** (770 deg C for iron) also removes magnetism.
* **Check** with a field indicator; residual field should be **below 3 gauss (0.3 mT)** .

15. Standardization and Calibration

* **System performance** verified with **test rings (Ketos ring)** having drilled holes at different depth, and test pieces with **known cracks** .
* **Field strength** checked using **pie gauge, QQI shims or gaussmeter** . Tangential field about **30 - 60 gauss (2.4 - 4.8 kA/m)** is the usual requirement.
* **Yoke lifting power test** before use (AC: 4.5 kg at maximum leg spacing, DC: 18 kg).
* **Particle concentration** checked by a centrifuge tube (settling test); **UV lamp intensity and white light level** with a calibrated meter; ammeter calibration

every six months.

16. Interpretation and Evaluation

Indication	Appearance
Surface crack	Sharp, well defined, tightly packed, continuous line
Subsurface flaw	Fuzzy, broad, less defined, usually with DC dry powder
Lamination, seam	Straight or curved lines
Non-relevant (magnetic writing, change of section)	Weak, broad, removed after demagnetisation

* ***Relevant indication*** - from a real discontinuity; ***non-relevant*** - due to design (sharp keyways, thread root); ***false*** - poor technique (particle trapped, rough weld toe).
* Evaluation as per ASTM E709, E1444, ASME Sec. V Art.7, IS 3703: cracks and linear indications over 1.5 mm usually rejected.

17. Effective Applications and Limitations of MPT

* ***Applications:*** weld inspection, castings and forgings, shafts, crank shaft, gears, bolts, railway axle, pressure vessel nozzles, aircraft landing gears, in-service fatigue crack checks.
* ***Advantages:*** fast, cheap, simple, high sensitivity to ***fine surface and near-surface cracks*** , shows flaw through thin coating, immediate result, portable, accept irregular shape.
* ***Limitations:*** ***only ferromagnetic materials*** ; flaws parallel to field not found; needs ***demagnetisation*** ; the surface must be reasonably clean; electric burns risk with prods; needs ***two directions*** for full coverage; poor depth information; large currents for big parts; paint and thick coating reduce sensitivity.

Comparison of Surface Methods

Property	LPT	MPT	ECT
Material	Non-porous	Ferromagnetic	Conductive
Flaws found	Surface-opening	Surface and slightly subsurface	Surface and near surface
Cost, speed	Low, slow	Low, fast	Medium, very fast
Contact needed	Yes (liquid)	Yes (particles)	No
Training	Low	Medium	High

Summary:

LPT - capillary action for non-porous; ECT - induced currents in conductors, change in impedance; MPT - leakage flux collects particles on ferromagnetic parts; MPT needs demagnetisation.

Important Questions

* Explain the steps of liquid penetrant testing with a neat flow chart.
* Discuss the principle and applications of eddy current testing.
* Explain skin effect and depth of penetration.
* Describe magnetisation techniques and the demagnetisation process in MPT.
* Compare LPT, MPT and ECT.